Electronic Engineering + Software Engineering

DIEGO ARMANDO PINTA CUASQUEN

From sensor to interface.

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PROFILE

I study Electronic Engineering and Software Engineering, and I work right where they meet: complete systems that start at a sensor or a microcontroller and end in an interface where data is seen and controlled.

EXPERIENCE

Embedded systems engineer and mobile developer, SISCAN Oct 2026 – present
Smart coffee drying: a hybrid solar-electric system for drying parchment coffee in Nariño, controlled by an ESP32.
Ongoing research: the control system and the mobile app are developed in parallel.
ESP32, Automatic control, Flutter, Dart
Case study Code: github.com/diarpicu2022-commits/siscan-movil

UI designer and frontend developer, NotaScan Oct 2026 – present
Academic grading system: AI reads the handwritten grade, the teacher verifies it and the system stores it.
Complete flow: AI detection, teacher verification and storage.
Tauri, React, TypeScript, Tailwind, Framer Motion
Case study Code: github.com/diarpicu2022-commits/notascan-escritorio

Game developer, Ecos de la Tierra Sep 2026 – present
Turn-based 2D pixel-art RPG about environmental awareness, exportable to PC and Android.
Six playable battles from a region selector, with five characters and the final boss.
Godot 4, GDScript, Python
Case study Code: github.com/diarpicu2022-commits/ecos-de-la-tierra

Android developer, CampusWatch / Agenda Sep 2026 – present
Personal Android planner for people who study and work, with a Wear OS module and encrypted data.
In development, with automated verification on an emulator.
Kotlin, Jetpack Compose, Wear OS, Room, SQLCipher
Case study Code: github.com/diarpicu2022-commits/agenda-android

Backend developer, HidroMon Oct 2026
REST API to monitor crop humidity with IoT sensors and threshold alerts.
API documented with an entity-relationship model and examples for every endpoint.
NestJS, TypeScript, Node.js
Case study Code: github.com/diarpicu2022-commits/hidromon-nestjs

Content designer and web developer, SETPasto+ Aug 2026
Redesign of Pasto's public transport portal, with a content and voice audit, in four languages.
Redesigned portal with audited content in four languages.
Node.js, Express, EJS, Tailwind
Case study Code: github.com/diarpicu2022-commits/Taller-Contenido--voz-y-internacionalizacion

Backend developer, EcoVolt May 2026
Solar energy monitoring and analysis to optimize panel performance.
Foundation for real-time solar production monitoring.
Python, FastAPI
Case study Code: github.com/diarpicu2022-commits/EcoVolt

Full-stack IoT developer, AgroPulse Mar 2026 – May 2026

End-to-end greenhouse monitoring: the node reads temperature, humidity and soil, and drives the pump, fan, heater and door.

Complete system across three repositories: firmware, backend and frontend.

`ESP32, C++, PlatformIO, Java 17, Spring Boot, Supabase, Docker, React, TypeScript`

Case study Code: github.com/diarpicu2022-commits/firmware-AgroPulse

Full-stack developer, Harmonia Apr 2026

Smart music player powered by AI.

Full-stack app with authentication, caching and AI integrated.

`Express, TypeScript, MongoDB, Redis, Groq, OAuth, React, Vite`

Case study Code: github.com/diarpicu2022-commits/Harmonia

Backend developer, Proxy IA Apr 2026

AI generation platform using the Proxy pattern to enforce per-plan quotas and rate limits.

The Proxy pattern keeps the business rule (quotas) apart from the model call.

`Java, Spring Boot, React, Docker`

Case study Code: github.com/diarpicu2022-commits/proxy-ai-backend

Game developer, Tower Defense Mar 2026

Tower defense game that uses the Decorator pattern to upgrade towers in real time.

A browser game where every upgrade is a visible decorator.

`Java, Canvas 2D, REST API`

Case study Code: github.com/diarpicu2022-commits/TowerDefenseDecorator

EDUCATION

Software Engineering · fifth semester Ongoing

Universidad Cooperativa de Colombia (UCC)

Software patterns, data structures, software design, interface design and web frameworks.

Electronic Engineering 2021 – 2025

Universidad de Nariño

Specialization track in Automation and Control. Electives in environmental variable measurement, electrical energy regulation in Colombia, and networks and telematic services.

SKILLS

Electronics	ESP32, Arduino, GPIO, ADC, DAC, PWM, I²C, OLED displays, Sensors and actuators
Engineering	MATLAB, Simulink, Automatic control, Modeling and simulation
Languages	C/C++, Java, TypeScript, JavaScript, Python, Kotlin, Dart, GDScript
Backend	Spring Boot, NestJS, Express, FastAPI
Frontend & apps	React, Next.js, Vite, Tailwind CSS, Flutter, Jetpack Compose, Tauri, Godot
Data	PostgreSQL (Supabase), MongoDB, Redis, SQLite, Room
Deployment & tools	Docker, Render, Vercel, Git & GitHub, PlatformIO, Wokwi

LANGUAGES

Spanish	Native
English	English proficiency requirement passed (Universidad de Nariño)